

WHY WE SHOULD STOP BEING SURPRISED THAT LIGHTLY-REGULATED MARKETS FAIL TO ACHIEVE THE SEC'S GOALS FOR MARKET QUALITY

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ABSTRACT

The stated goals of the SEC are to protect investors, maintain orderly markets and facilitate capital formation. These goals can be achieved with very light regulation if, as assumed by traditional economic theory, investors process information costlessly and protect themselves from informational disadvantages, and firms optimally balance the costs and benefits of committing to make their reports reliable. A growing body of research demonstrates that light regulation fails to achieve the SEC's goals, because investors find information processing costly and fail to protect themselves. After reviewing theory and prior evidence, I discuss new lessons learned from Jiang, Petroni and Wang (2015), who show that PinkSheets® reduced the liquidity of firms with low reporting quality and increased the liquidity of firms with high reporting quality, merely by highlighting the quality of their listed firms' disclosure. While the Pink Sheets® innovation might have occurred through many causal channels, all of them entail a violation of costless processing and self-protection, and lead to the conclusion that this lightly regulated market did not initially meet the stated goals of the SEC. I conclude by arguing that markets can achieve the SEC's goals only if they exhibit a particularly strong version of "dynamic" market efficiency, which requires that each individual trade on the path to even incomplete revelation occur at the then-optimal price. Because dynamic efficiency is unlikely, we should stop being surprised to see yet more evidence that lightly-regulated markets fall short on key dimensions. Instead, we should use our well-developed understanding of market inefficiency to guide regulation.

1. INTRODUCTION

Many accounting scholars have studied the impact of regulatory changes in markets that are already subject to strong regulations.¹ Jiang, Petroni and Wang (2016, hereafter 'JPW') depart from this familiar path by examining the impact of a change imposed by a private exchange, Pink Sheets®, a market subject to only very weak regulatory oversight.

¹ See the 2010 Conference issue of the Journal of Accounting and Economics (Vol 49, Issues 1-2 for some reviews of this literature.)

Specifically, in 2007, Pink Sheets® altered their own presentation of firm’s listings by classifying firms according to the quality of their disclosures (Current Information, Limited Information, and No Information) and making those classifications strikingly salient with colorful graphics (a version of a Pink Sheets® logo, a red yield sign, and a red stop sign, respectively). JPW find that this innovation caused an increase in market liquidity for shares of firms classified as providing Current Information, and a decrease in market liquidity for shares of firms classified as providing No Information. They also find that investors seemed to anticipate this effect, with prices dropping for the former group, over the days in which the upcoming change in presentation was announced.

In Section 2, I place the paper in the context of other archival studies that examine lightly-regulated markets, and argue that such studies provide useful insights into an idealized but never-observed counterfactual that frequently appears in regulatory discussions: the fully libertarian market free of all regulations, in which traders must protect themselves against both misleading reporters and better-informed traders. In Section 3, I argue that JPW’s results provide additional evidence that lightly regulated markets do not achieve the goals of the SEC because information processing is not costless for at least some investors, and investors do not protect themselves as effectively as traditional economic theory would predict. In Section 4, I further argue that the results are consistent with—and in fact totally unsurprising in light of prior evidence of these two investor failures, and that these failures are robust enough to guide regulators.

In Section 5, I discuss the challenges in determining precisely why the innovation had the effects that it did. Because Pink Sheets® intended their innovation in search of profit, not clean scholarly inference, they confounded many possible explanations. Future research could gather new data from laboratory experiments, field experiments, field studies and surveys to clarify the causal explanation. In Section 6, I conclude by emphasizing that ‘lightly regulated markets don’t achieve the goals of the SEC’ is a positive claim about what happens when reporting standards are left to the discretion of managers and exchanges. This claim does not imply that heavily regulated markets (such as those for large public firms in the US) would achieve the SEC’s goals more effectively with yet more regulation, nor does my discussion establish that the SEC’s goals are normatively desirable ones to achieve. Those who disagree with either of these claims can validly argue for less regulation, as long as they do not rely on the claim that investors protect themselves well enough to compensate for the changes they recommend.

2. THEORY AND PRIOR EVIDENCE ON LIGHTLY REGULATED AND UNREGULATED MARKETS

How would markets behave in the absence of any regulation? Specifically, how well would they achieve the mission of the U.S. Securities and Exchange Commission, which is “to protect investors, maintain fair, orderly, and efficient markets, and facilitate capital formation”?² Traditional economic theory allows us to answer this question by imposing a number of assumptions, like individual rationality, rational expectations and risk aversion, but I will focus on three that are particularly important: the costs of processing publicly

² <http://www.sec.gov/about/whatwedo.shtml>

available information are trivially small (costless processing); traders protect themselves against others with superior information (self-protection); and managers invest optimally in costly technologies, like auditing, that commit themselves to restricting their reporting flexibility, with the restriction increasing with cost (reporting commitment).

The twin assumptions of costless processing and self-protection guarantee that investors are protected, and markets are mostly fair and efficient. Investors will not be duped by managers, because they simply won't rely on reports that are not made reliable by some type of auditing technology. Less-informed investors will not lose money by trading with more-informed investors, because self-protection assures that they will avoid trading in ways that expose themselves to such losses. And markets will be efficient (at least with respect to public reports) because all investors process those reports completely and incorporate them into their trading decisions.

Reporting commitments are necessary for markets to facilitate capital formation and be orderly (which I define as being liquid enough to allow investors to open and close positions at low cost). If managers have no way to limit reporting flexibility, investors who lack access to other reliable information will protect themselves by simply not investing. This will cause market liquidity to plummet, since so little money is invested in the first place, forcing sellers to pay steep prices to find willing buyers. However, managers can solve these problems by investing in auditing and other technologies to make their reports more reliable. This allows them to raise capital even from skeptical investors, and helps markets be orderly because there are non-informational reasons for trading shares, such as the need to share risk and fulfill liquidity needs, which allows buyers and sellers to find ready counterparties at low cost.

Traditional economic assumptions allow an elegant case against mandatory reporting regulations, because costless processing, self-protection and reporting commitment not only allow markets to achieve all of the elements of the SEC's mission, but they also optimally balance the costs and benefits of audit technology: firms will invest in audit technology only to the extent that its cost are outweighed by the benefits of capital formation.

Unfortunately, this elegant argument against reporting regulation seems to work poorly in the real world, and JPW provide yet more evidence of its inadequacy. Before discussing JPW's results, it is useful to note how difficult it is to study this question using markets already subject to strong regulation. There are too many studies to count that address the impact of regulatory changes like Sarbanes-Oxley, new reporting standards and new auditing standards. But all of these changes start from a baseline of extensive regulation. To the extent that existing regulations already create market imperfections, such research runs headfirst into the 'Theory of the Second Best' (Lipsey and Lancaster 1956): one cannot use economic theory to prove that eliminating one imperfection will improve welfare as long as there are other imperfections (because imperfections may offset one another). Thus, we can draw few conclusions about the efficacy of unregulated markets from pro-regulatory evidence that Sarbanes-Oxley lowers required rates of return by lowering reporting flexibility (Bloomfield, M. 2015), or anti-regulatory evidence that its excessive costs will cause firms to flee the market (Leuz 2007).

Fortunately, some researchers have devoted their attention to markets that are very lightly regulated or not regulated at all. Gerakos, Lang and Maffett (2013) provide evidence from the London Stock Exchange's Alternative Investment Market (AIM). Firms that list on these markets are not required by regulation or exchange rules to meet any particular standards for reporting or auditing; instead, they nominate an advisor who determines appropriate practices and attests that the firm adheres to them. But the empirical evidence suggests that such markets do dupe investors; returns on such firms are well below returns of comparable but regulated firms, and Gerakos et al. (2013) do a persuasive job of showing that the low returns are not driven by firm's choices to list themselves on AIM or investors' rational desire to earn enormous returns on rare successes.

Bloomfield and Cho (2011) provide similar evidence from completely unregulated markets: the SLCapEx exchange that allowed participants in the virtual world Second Life to raise approximately \$600,000 in US dollars from other participants. Those raising capital purported to do so to support various business operations, like developing software and other digital assets, but it is hard to know their true intentions, or how they actually used the money they raised, because there were no regulations governing their reporting activities, no legal requirements that separated their individual and business finances, and none of those raising money invested in technologies that would improve the reliability of their reports or the credibility of their commitments to pay appropriate dividends. Nevertheless, they raised a great deal of money, only 6% of which was ever returned to investors.

3. PRE-INNOVATION PINK SHEETS® MARKETS DID NOT ACHIEVE THE SEC'S GOALS

JPW provide new evidence that investors don't protect themselves as effectively as predicted by traditional economic theory, and that lightly regulated markets are therefore unlikely to achieve the SEC's goals of protecting investors and making markets "fair." To be clear, this is my argument, not theirs. JPW claim only that their results show "how a private intermediary innovation can shape a market-wide phenomenon without costly regulation" and that "intermediaries can affect investor behavior through simple labels and/or graphics that draw investors' attention to a firm's existing disclosure practice". Because JPW focus on liquidity, rather than investor losses, the argument is more complex than in Bloomfield and Cho (2011) and Gerakos et al. (2013). In this section, I elaborate on the reasons for my conclusion, by arguing that information processing is not costless for Pink Sheets® investors, and that at least some Pink Sheets® investors do not protect themselves from deceptive managers and better-informed investors.

Information Processing is not Costless for some Pink Sheets® Investors

My interpretation of JPW's results hinges on my judgment that the Pink Sheets® innovation provides no new information to investors, nor does it reduce their information processing costs in any manner that would be consistent with the traditional assumption of costless processing. The Pink Sheets® innovation was a simple one: the information intermediary began examining the disclosures that firms already provided on the website, and determined whether they are "foreign firms that are listed on 'qualified exchanges' [or] domestic firms that file continuous financial statements to the SEC or other regulators or

make adequate ‘filings publicly available through the Pink Sheets News Service’” [JPW, p8]. If so, they were classified as “Current Information”. It takes investors little time to make this classification themselves, and all of the information is readily available. The classification of “No Information” requires only occasional investigation, because it requires information to be at least six months out of date. One could argue that investors did not realize the importance of this assessment or lacked the ability to understand the information required to make these assessments, but such arguments would hardly be consistent with the assumption of costless processing.

If one accepts my judgment on the nature of the Pink Sheets® innovation, traditional economic theory predicts it would have no effect on market behavior at all. But JPW provide strong evidence of a particular effect: it causes liquidity for Current Information firms to rise and No Information firms to fall. They examine four distinct measures of liquidity: bid-ask spreads, the sensitivity of price to trading volume, the number of days on which trades occur, and trading volume. They also calculate a liquidity factor that combines all four of the measures. This breadth of measures establishes the robustness of the effect.

To rule out omitted variables, JPW compare pre- and post-innovation liquidity to comparable firms that are traded on other exchanges under similar regulations, but are not subject to the new classification scheme (those traded on the OTC Bulletin Board). To rule out any time trend, JPW perform a placebo test by moving the innovation event a few months earlier and document no liquidity changes after the pseudo-event. Because Pink Sheets® implement the classification schemes based on firms’ existing disclosure and their implementation is rather swift, self-selection and reverse-causality are less likely. To rule out that firms’ change of disclosure drive the liquidity changes, JPW also randomly select 200 firms and compare their disclosure policy during the pre- and post-innovation periods. For majority of these firms, there are no discernable changes in their disclosure levels. Thus, the study does a good job of establishing causal identification, which Angrist and Pischke (2008) characterize as approximating the ideal of a controlled experiment in which the objects of study (in this case, firms) are randomly assigned to treatments (in this case, the classification system).

Pink Sheets® Investors do not Protect Themselves

My interpretation of JPW’s results also hinges on my interpretation of the directional changes in liquidity. I derive my interpretation from two sets of assumptions. The first set of assumptions describes reporting flexibility and investor information. I assume that managers distort their reports up or down (depending on their particular goals at the time), and do so more when they have more reporting flexibility. To match JPW, we can assume low, medium and high levels of flexibility to Pink Sheets® current, limited and no information classifications. There are three types of traders: less-informed traders who know only what is publicly reported, more-informed traders who know the public reports and the distortion, and noise traders who buy or sell at any price to satisfy their liquidity needs (e.g., investing a bonus, withdrawing to pay for college). Finally, I assume that the Pink Sheets® innovation allows both noise traders and less-informed traders to know whether a firm’s reporting flexibility is low, moderate or high; before the innovation, they

know only the proportions of the three types, but not whether any particular firm has a particular level of flexibility.

The second set of assumptions describes the nature of trade. I assume that the market can be modeled as a 'clearinghouse', in which trade is divided into many periods. In each period, every trader submits a complete demand schedule that specifies how many shares they will buy or sell at every possible price. The market clears at the price that equates supply and total demand. Trade does not actually work this way; it is better described as a continuous electronic limit order book (ELOB), in which traders enter offers to buy or sell shares at the limit price they specify, or take others' offers at limit prices specified by others. However, the ELOB market model is notoriously intractable. In contrast, the clearinghouse paradigm maps directly into most familiar workhorse of traditional market models, the noisy rational expectations market, and is sufficient to capture two key aspects of market orderliness: price pressure and volume. Price pressure is the extent to which the market price will move if a hypothetical investor attempts to buy or sell N additional shares, which is determined by the average slope of the aggregate demand schedule. In a highly liquid market, the aggregate demand schedule is nearly vertical, so that you could buy or sell a large number of shares without affecting the price at all, since the market will be willing to sell or buy many more shares at prices slightly above or below the market price.

We now have the apparatus to establish that investors were not protecting themselves against biased disclosure before the Pink Sheets® innovation, and that as a result the markets were not as fair as after the innovation (and therefore that a regulation mandating such categorization would have better achieved the SEC's goals). I divide this argument into two parts. First, I assume that markets are perfectly rational. In this case, liquidity traders transfer wealth to less-informed and more-informed traders, and the market as a whole transfers wealth to the managers. The liquidity traders transfer wealth to other investors because they have no choice over their demand, and no information that allows them to concentrate their trades in the current-information markets: they buy or sell the number of shares demanded by their random liquidity shocks, split equally across all three markets. The less-informed investors also have no information to tell them which stocks to trade, but they have more flexibility in trading than the liquidity traders. They can therefore enter trades equal to their reservation price, and expect to buy at low prices when liquidity traders are selling, and sell at high prices when liquidity traders are buying. However, the less-informed need to be extremely cautious, trading only a small number of shares, because they will systematically lose money to the more informed, who will set wiser reservation prices, selling when the firm has inflated reported value and buying when the firm has deflated it. The less-informed can still make money overall, however, because the more-informed traders are not willing to bear an infinite amount of risk, and therefore prices reflect aggregate information only incompletely. Thus, even with traders who are perfectly rational (but face significant information processing costs), the market is not particularly fair to liquidity traders.

Second, I argue that the markets do not suggest a very high degree of rationality. One reason is that, as discussed above, it is not very rational for traders to act as if information processing is costly. Even conceding costly processing as rational, the magnitude of JPW's effects are hard to reconcile with self-protection. It is hard to believe that very many

liquidity traders choose to resolve their liquidity needs in Pink Sheets® markets, when they could instead access highly regulated markets that would protect them more effectively. This in turn implies that there would be very few rational less-informed traders, because with few liquidity traders to exploit, they have little opportunity to profit, and will consistently lose money to more-informed traders. This suggests that most of trade in the pre-innovation markets is driven by less-informed traders who fail to protect themselves from more-informed traders. Finally, it is worth noting that after the innovation, the no-information markets are not very orderly (due to low liquidity), and are therefore unlikely to facilitate capital formation. Thus, it is hard to reconcile JPW's results with the elegant claim that traditional economic assumptions allow markets to achieve the SEC's goals, even when firms can choose their own level of auditing technology.

4. FINDINGS LIKE JPW'S SHOULD NO LONGER SURPRISE US, BUT INSTEAD SHOULD SERVE AS A BASIS FOR POSITIVE REGULATORY CLAIMS

In the previous section, I argued that JPW's results suggest that lightly regulated markets did not satisfy the goals of the SEC, because investors process information as if it is costly, and fail to protect themselves from informational disadvantage. In this section, I argue that the particular forms of inefficiency implied by JPW's results should no longer be considered surprising, and that we have a model of imperfectly efficient markets that is sound enough, empirically and theoretically, to serve as a basis of regulation.

The particular forms of inefficiency implied by JPW's results should no longer be surprising

Evidence of costly processing is familiar in accounting, and discussions of the phenomenon increasingly use the framework of the Incomplete Revelation Hypothesis (Bloomfield 2002). The IRH assumes that information processing is costly, even when data are publicly available, and that noise in markets presents information from being fully impounded in price unless it is processed by all traders. The first assumption simply reflects common sense, embodied in the familiar term 'information overload'. The second assumption reflects a large body of experimental studies (e.g., Bloomfield and Libby 1996). Taken together, along with other assumptions regarding what public information is costly to process, the IRH provides a coherent theory for interpreting many archival studies that are hard to reconcile with the Efficient Markets Hypothesis. Among the most famous of these is the 'accrual anomaly': markets underreact to the transitory nature of accruals, which are relatively hard to process (Sloan 1996).

Accountants have devoted less attention to investors' failure to protect themselves. However, this inefficiency should not be surprising either. The most fundamental failure of self-protection is the winner's curse: the winner of an auction typically loses money. The winner's curse has been established by hundreds of laboratory experiments as one of the most robust results in experimental economics, and one that is very hard to extinguish. It was inspired by evidence from the field (winners of oil and gas plots tended to lose money),

and has been subsequently confirmed in a variety of experiments (Capen, Clapp, and Campbell, 1971; Thaler, 1988). Bloomfield, Libby and Nelson (1999) demonstrate that the winner's curse generalizes to accounting settings by showing that less-informed investors systematically transfer wealth to more informed investors, unless they are given a heavy-handed prompt to consider their informational disadvantages. Hales (2009) demonstrates a similar result in a context more closely tied to clearinghouse markets.

To further clarify why certain forms of inefficiency should no longer surprise us, it is useful to consider a hierarchy of efficiencies that markets can exhibit, all with respect to information set X, which most commonly will refer to information that can be extracted from public disclosures.

- **No-profit efficiency** (with respect to information set X) demands that no hypothetical investor holding only X could enter the market and earn abnormal returns, after accounting for transactions costs and information processing costs. No-profit efficiency is relatively easy to achieve after a few days of trading, because the pursuit of abnormal returns is the engine that drives markets toward efficient outcomes. If markets cannot achieve no-profit efficiency after a few days of trading, they will not achieve any other definition either. However, it is worth emphasizing that no-profit efficiency is a modest benchmark. It is easily achieved in a market with high transactions costs, but that is hardly a cause for celebration. Moreover, a market that achieves no-profit efficiency need not achieve any of the goals of the SEC, as will be clear once we define the other forms of efficiency.
- **Fair-price efficiency** demands that prices are the same as they would be if all investors held information set X. This form of efficiency achieves the SEC's goal of fairness, and protects less-informed investors from more-informed investors, because market price already fully reflects X, and it is therefore *as if* less-informed investors know X as well. However, markets that achieve no-profit efficiency will still typically not attain fair-price efficiency, because the definition of the latter ignores both transaction costs and information processing costs. As a result, the market price will stall before achieving fair-price efficiency, because no one is willing to process information that cannot be used profitably in trading. However, prices are not fair-price efficient because if the information set X were somehow given to a hypothetical new trader, that trader would cause the price to reflect X more completely.
- **Right-price efficiency** demands that prices accurately reflect the implications of X. No-profit efficiency does not imply right-price efficiency for the same reasons it does not imply fair-price efficiency. Fair-price efficiency does not imply right-price efficiency, since all investors might misconstrue the meaning of X.³ By defining X as

³It is difficult to imagine scenarios in which a market systematically achieves right-price efficiency without achieving fair-price efficiency, but this could happen in special circumstances. For example, if all investors

information about misreporting, we can see that right-price efficiency comes closer to achieving the SEC goals, because it guarantees that investors as a class are protected from misreporting.

- **Dynamic efficiency** demands that every order entered into or taken from the market is an optimal choice for the trader doing so, even before the market fully impounds X into market price. Dynamically efficient markets protects investors far more effectively than right-price efficient markets, because it eliminates the possibility that less-informed investors lose money to more-informed investors during trades immediately after X is disclosed. However, dynamic efficiency is almost impossible to achieve, even in extremely simple settings. The winners' curse is the most obvious example of failure, as is the routine evidence that less-informed traders systematically lose money in laboratory markets. (Their trades are clearly suboptimal, because they would have earned more by not trading.)

The most natural interpretation of JPW's results is that pre-innovation Pink Sheets® markets did not achieve dynamic efficiency. We can see this because less-informed investors stopped posting orders for no-information stocks once the cost of processing was lowered by the innovation. Given the hierarchy of efficiencies listed above, this should hardly be surprising.

It seems unlikely that pre-innovation Pink Sheets® markets achieved right-price or fair-price efficiency, but JPW provide only indirect evidence on this point, by showing that returns for no-information stocks fell, and (for one event) returns for current information stocks rose, when steps toward the innovation were announced. I interpret these moves as indicating that the more-informed traders understood that the markets failed to achieve fair-price and right-price efficiency, and that prices for no-information stocks would drop when the innovation provided less-informed traders with more information.

We have a model of inefficiency that is sound enough to serve as a basis of regulation.

Researchers who continue to be surprised by results like those reported by JPW, despite their theoretical grounding and repeated empirical support, often pair their surprise with a misunderstanding of the alternative to market efficiency. In particular, they tend to think the alternative to efficiency is any and all *ad hoc* forms of inefficiency, as opposed to the narrow forms that are indicated by relaxing the assumptions of costless processing and self-protection. As one example, consider the following claim by Kothari, Ramanna and Skinner (2010, p279):

1. Market inefficiency is not an equilibrium theory: Unlike the efficient market hypothesis, which describes a capital market pricing equilibrium, behavioral theories

overestimate the reliability of X, the market would be right-price efficient if exactly the right portion of traders held X, while overreacting if all of them did.

about market inefficiency describe transient pricing, i.e., states that are not expected to persist in perfect market conditions. Moreover, there is no behavioral theory to describe the relation of accounting information to stock market prices in an equilibrium of market inefficiency. Absent an equilibrium theory of market inefficiency, regulation that assumes inefficiency has no natural starting point, and more importantly, no framework to guide markets back to efficiency. In other words, if GAAP is designed assuming market inefficiency, then it is unclear how such a GAAP would lead to an equilibrium state of market efficiency. Without a framework to understand the origin and persistence of irrational pricing, several important questions arise: Would inefficiency persist no matter what is the design of GAAP? Or worse, can inefficiency be exacerbated through poorly understood and thus poorly designed regulation?

2. Practical difficulties with the market inefficiency assumption: As a practical matter, even if standard setters were to embrace inefficiency as the maintained assumption, we doubt market inefficiency has the potential to guide them in deciding on a suitable GAAP. What behavioral assumption should be assumed and therefore what form of inefficiency should be assumed? Should we assume prices over-react or under-react? Do they initially under-react, but then over-react if a firm reports a sequence of good news or a sequence of bad news, which triggers representativeness bias? How long should such a sequence be before under-reaction morphs into over-reaction on the part of investors? What should we assume with respect to arbitrage opportunities and the likely degree of success of arbitrageurs?

Surely it would be difficult to develop standards based on a non-specific theory of market inefficiency. Echoing the *Anna Karenina* principle that “Happy families are all alike; every unhappy family is unhappy in its own way,” one might note that efficient markets are all similar in that they meet a common set of requirements (such as fully revealing prices and no arbitrage opportunities), but that there are many ways to deviate from these requirements.

However, this is a misapplication of the *Anna Karenina* principle, which is appropriate only when one cannot narrow the possible causes of flaws. JPW show (yet again) that inefficiencies are fairly well captured by the twin failures of costly processing and self-protection. These failures are equilibrium outcomes, both theoretically, because they can be modeled as no-profit efficient outcomes under relaxed assumptions, and empirically, because they explain reliable deviations from efficiency in the lab and in real markets (as JPW show). Moreover, they provide natural guidance for regulators: regulators can achieve the SEC’s goals by limiting reporting flexibility when possible, and ensure that investors can learn about reporting flexibility at low cost, as Pink Sheets® has done. Given that regulators and private bodies like Pink Sheets® take exactly such actions, I would suggest that my approach describes how policy makers and practitioners actually behave, and does so more accurately than the approach recommended by Kothari, Ramanna and Skinner.

5. EXPERIMENTAL WORK CAN CLARIFY WHY THE PINK SHEETS® INNOVATION WORKED

JPW provide strong evidence that the Pink Sheets® innovation had strong causal effects. As discussed in Section 3, they establish three of the four elements of causal attribution identified in Bloomfield, Nelson and Soltes (2016): that causation runs in the correct direction (the innovation caused changes in liquidity, rather than the other way around); that the association is moderated as theory would predict (by reporting quality); and that the association isn't driven by variables other than the specified cause (established through controls and the careful design).

However, JPW do not establish the fourth element of causal attribution: that association is driven by an appropriately narrow definition of the constructs specified. In fact, it is difficult for them to specify any particular constructs at all, because the innovation was not intended as a scientific experiment. Instead, Pink Sheets® used a 'shotgun approach' to its innovation that enhanced the likelihood of improving market quality, but confounds a variety of causal forces that might account for its effects. Here are six forces that might play a role in driving JPW's results, all of which suggest natural directions for experimental research:

1. *Reduced Costs of Information Processing.* The Pink Sheets® innovation may have changed liquidity because the new classifications lowered the cost of processing information about reporting quality.
2. *Increased Salience of Reporting Quality.* The Pink Sheets® innovation could have made reporting quality more salient to investors. A wealth of research in psychology shows the power of visual inputs on human attention and judgment. Pink Sheets® choose particularly familiar symbols (e.g., stop signs) that tap into existing heuristics (stop signs mean stop!). Thus, these effects could be quite strong.
3. *Increased Salience of Informational Disadvantages.* Experimental studies suggest that people do a better job of protecting themselves against informational disadvantages when they are cautioned about them, even though the caution contains no information itself (Bloomfield, Libby, and Nelson 1999; Hales 2009).
4. *Pragmatic Effects.* Grice's pragmatic theory of language contends that people expect communicators to highlight relevant information and downplay irrelevant information (Bloomfield 2012). Under this theory, it would be natural for investors to infer that Pink Sheets® is communicating that disclosure quality is more important than they had previously thought.
5. *Demand Effects.* Experimenters must always be concerned about the possibility that their manipulations alter behavior because subjects are attempting to please them, or are simply exhibiting a natural tendency to do what is expected of them. Investors are likely to interpret the innovation as indicating the preferences or expectations of Pink Sheets® that traders avoid trading firms labeled with warnings.
6. *Speculative Effects.* Investors might have altered their behavior in response to the Pink Sheets® innovation not because it changed their own beliefs, but because they expect it to change other traders' beliefs or actions. It is well known that liquidity attracts

liquidity. Thus, traders who expect other traders to migrate toward full-information stocks will do so themselves, even if they think other traders' actions are irrational.

Each of these explanations has subtly different implications for the nature of inefficiency before and after the PinkSheets® innovation, but future experimental work would be needed in order to evaluate their explanatory power.

6. CONCLUSION

JPW provide useful new evidence that lightly-regulated markets do not achieve the SEC's goals regarding market quality. Many of their findings are surprising to proponents of efficient markets theory, but familiar and unsurprising to those who accept that investors neither process information costlessly nor protect themselves perfectly against informational disadvantages. While the PinkSheets® innovation does not allow us to determine the exact reason for its effects, it still provides strong evidence that those effects are causal, not merely associational. The innovation can therefore provide a useful guide for regulators who wish to improve market quality, and are receptive to well-specified alternatives to the Efficient Markets Hypothesis.

Finally, it is worth emphasizing that 'lightly regulated markets don't achieve the goals of the SEC' is a positive claim about what happens when reporting standards are left primarily to the discretion of managers and exchanges. I am effectively arguing that the relationship between market quality and regulation exhibits an inverted-U shape similar to the Laffer curve's relationship between government revenue and marginal tax rates. In both cases, the extreme choices (no regulation and massive regulation for markets, 0 and 100% marginal rates for taxes) can be improved upon by shifting toward more moderate choices. My claim therefore does not imply that heavily regulated markets, like those for large publicly traded firms in the US, are too lightly regulated. Note also that I do not claim that the SEC's goals are normatively desirable ones. Those who place high value on liberty for its own sake can argue that less regulation is always better because the intrinsic value of liberty outweighs the value of investor protection, orderly markets and capital formation. However, one cannot claim that less regulation *always* achieves the SEC's goals more effectively because investors process information costlessly and will protect themselves.

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